

Review: Minimax Lower Bound v2 (Hellinger Distance Proof)

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Scope: Full mathematical verification of all inequalities, formulas, and derivations. Numerical cross-checking of all claimed relationships.

1. Overall Verdict

The proof correctly fixes the three fatal issues from v1 (Slud's inequality, chi-squared direction, F1 algebra), and the core Hellinger tensorization approach is sound. **However, two new fatal errors have been introduced in the F1 conversion section** (Section 7), and a significant formula error appears in the $K > 2$ section (Section 8.3).

Bottom line: The testing error lower bound (Part a) is correct. The F1 lower bound (Part b) has two algebraic errors that invalidate its stated constant. The rate optimality claim (Part c) remains correct because the exponential rate $\exp(-2M\Delta^2)$ is preserved under the corrected analysis.

2. Issues Found

Issue 1 [FATAL]: Section 7.3 – Backwards inequality in F1 simplification

Location: Lines 423-427

Claim:

$$(1-\eta) \cdot (\rho^M/4) / (2\eta + (1-\eta) \cdot (\rho^M/4)) \geq (1-\eta) \cdot \rho^M / (8\eta)$$

What is wrong: The inequality is backwards. Since $2\eta + (1-\eta) \cdot (\rho^M/4) > 2\eta$ for any positive ρ^M , the left-hand side has a larger denominator with the same numerator, making it strictly smaller. The correct inequality is $\leq$, not $\geq$.

Verification ($\eta = 0.3$, $\rho^M = 0.1$):

$$\text{LHS} = 0.7 \cdot 0.025 / (0.6 + 0.7 \cdot 0.025) = 0.02834$$

$$\text{RHS} = 0.7 \cdot 0.1 / (8 \cdot 0.3) = 0.02917$$

LHS < RHS, so LHS $\geq$ RHS is FALSE.

Consequence: The simplified bound $1\text{-F1} \geq \rho^M / (16\eta)$ is not validly derived. The correct inequality direction gives an upper bound LHS $\leq$ RHS, which is useless for lower-bounding 1-F1.

How to fix: The exact bound must be kept in its unsimplified form:

$$1\text{-F1} \geq (1-\eta) \cdot (\rho^M/4) / (2\eta + (1-\eta) \cdot (\rho^M/4))$$

Or a correct simplification with a multiplicative correction:

$$1\text{-F1} \geq (1-\eta) \cdot \rho^M / (8\eta) \cdot (1 - (1-\eta) \cdot \rho^M / (4\eta))$$

(using $1/(1+x) \geq 1-x$ for $x \geq 0$)

Issue 2 [FATAL]: Section 7.5 – Wrong case selected as tighter bound

Location: Lines 439-467

Claim:

$$\text{eps}/2 \geq (1-\text{eta})*\text{eps}/(2\text{eta}+(1-\text{eta})*\text{eps}) \text{ for } \text{eta} \leq 1/2$$

And consequently:

$$\min(\text{eps}/(2-\text{eps}), (1-\text{eta})*\text{eps}/(2\text{eta}+(1-\text{eta})*\text{eps})) = (1-\text{eta})*\text{eps}/(2\text{eta}+(1-\text{eta})*\text{eps})$$

What is wrong: The inequality direction is reversed for $\text{eta} < 1/2$ and small eps (the regime of interest). The proof attempts to verify this by cross-multiplication, claiming:

$$2\text{eta} + (1-\text{eta})*\text{eps} \geq 2*(1-\text{eta}) \text{ follows from } \text{eps} \geq 0, \text{ eta} \geq 0.$$

But the correct cross-multiplication is:

$$\begin{aligned} \text{eps}/2 &\geq (1-\text{eta})*\text{eps}/(2\text{eta}+(1-\text{eta})*\text{eps}) \\ \Rightarrow 2\text{eta} + (1-\text{eta})*\text{eps} &\geq 2*(1-\text{eta}) \\ \Rightarrow (1-\text{eta})*\text{eps} &\geq 2 - 4\text{eta} \end{aligned}$$

For $\text{eta} = 0.3$: requires $\text{eps} \geq 0.8/0.7 \approx 1.14$. Impossible since $\text{eps} \leq 1$ (and typically $\text{eps} \ll 1$). For $\text{eta} = 0.4$: requires $\text{eps} \geq 0.4/0.6 \approx 0.667$. Possible but atypical – ρ^M would need to be ≥ 2.67 , impossible since $\rho \leq 1$. For $\text{eta} = 0.49$: requires $\text{eps} \geq 0.04/0.51 \approx 0.078$. Possible for some parameters. For $\text{eta} = 0.5$: requires $\text{eps} \geq 0$. True.

So the claim only holds when $(1-\text{eta})*\text{eps} \geq 2-4\text{eta}$, i.e., when $\text{eta} \geq$

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$1/(2-\epsilon/(1-\epsilon))$ approximately. For most of the parameter space (especially small ϵ and small-to-moderate η), the claim is FALSE.

Verification ($\eta = 0.3$, $\epsilon = 0.1$):

$$\epsilon/2 = 0.05$$

$$(1-\eta)*\epsilon/(2\eta+(1-\eta)*\epsilon) = 0.7*0.1/0.67 \approx 0.104$$

So $\epsilon/2 < \text{case2_bound}$, not $\geq$.

The correct minimum is: $\min(\epsilon/(2-\epsilon), (1-\eta)*\epsilon/(2\eta+(1-\eta)*\epsilon))$
= $\epsilon/(2-\epsilon)$ (Case 1)

for all $\eta < 1/2$ and small ϵ .

Consequence: The claimed universal bound $1-F_1 \geq (1-\eta)*\epsilon/(2\eta+(1-\eta)*\epsilon)$ is invalid. The correct universal bound uses Case 1:

$$1-F_1 \geq \epsilon/(2-\epsilon) = \rho^M/(8-\rho^M)$$

which is approximately $\rho^M/8$ (not $\rho^M/(16\eta)$).

Comparison of bounds:

$$\eta=0.1: \text{correct}=\rho^M/8, \text{claimed}=\rho^M/1.6 \quad (\text{claimed } 5x \text{ too large})$$

$$\eta=0.3: \text{correct}=\rho^M/8, \text{claimed}=\rho^M/4.8 \quad (\text{claimed } 1.67x \text{ too large})$$

$$\eta=0.5: \text{correct}=\rho^M/8, \text{claimed}=\rho^M/8 \quad (\text{match at } \eta=0.5)$$

How to fix: Use the Case 1 bound:

$$1-F_1 \geq \epsilon/(2-\epsilon) = (\rho^M/4)/(2-\rho^M/4) = \rho^M/(8-\rho^M)$$

For the simplified form (small ρ^M):

$$1-F_1 \geq \rho^M/8$$

Issue 3 [MAJOR]: Section 8.3 – Incorrect Hellinger affinity formula for $K > 2$

Location: Line 511

Claim:

$$\text{rho_K} = 2 * \text{sqrt}(\text{mu_s} * (1 - \text{mu_s}/(K-1)))$$

What is wrong: This formula is only correct for $K=2$. The general Hellinger affinity for $\text{Bernoulli}(p)$ vs $\text{Bernoulli}(q)$ is:

$$\text{rho} = \text{sqrt}(p*q) + \text{sqrt}((1-p)*(1-q))$$

The proof's formula $2 * \text{sqrt}(p*q)$ only matches the general formula when $q = 1-p$, i.e., when $K=2$ (since $1 - \text{mu_s}/(K-1) = 1 - \text{mu_s} = 1 - p$ only when $K=2$).

For $K > 2$, the correct formula is:

$$\text{rho_K} = \text{sqrt}(\text{mu_s} * (1 - \text{mu_s}/(K-1))) + \text{sqrt}((1-\text{mu_s}) * (\text{mu_s}/(K-1)))$$

Verification ($\text{mu_s} = 0.3$, $K = 3$):

$$\text{correct: } \text{sqrt}(0.3*0.85) + \text{sqrt}(0.7*0.15) = 0.505 + 0.324 = 0.829$$

$$\text{wrong: } 2 * \text{sqrt}(0.3*0.85) = 2 * 0.505 = 1.010$$

The wrong formula gives $\text{rho} > 1$ for many parameter values (e.g., $\text{mu}=0.3$, $K=3$: $\text{rho}=1.010$; $\text{mu}=0.2$, $K=5$: $\text{rho}=0.872$). While $\text{rho} \leq 1$ should always hold for Hellinger affinity. The correct formula always satisfies $\text{rho} \leq 1$.

Consequence: The numerical values and asymptotic expressions for $K > 2$ are wrong. However, the **conclusion** that $K=2$ is the hardest case is still correct (verified with the correct formula – $\text{rho_K} < \text{rho_2}$ for $K > 2$, giving

a weaker bound, so $K=2$ gives the strongest valid lower bound for all $K \geq 2$).

How to fix: Replace the formula with the correct expression:

$$\text{rho_K} = \sqrt{\mu_s * (1 - \mu_s / (K-1))} + \sqrt{(1-\mu_s) * (\mu_s / (K-1))}$$

And note that $\text{rho_K} < \text{rho_2} = 2 * \sqrt{\mu_s * (1-\mu_s)}$ for all $K > 2$, which can be verified by squaring both sides.

Issue 4 [MAJOR]: Section 11 – Chernoff exactness claim fails for $K > 2$

Location: Lines 676-681

Claim:

$$-\log(\text{rho}) = \text{KL}(1/2 \parallel \mu_s) = \text{Chernoff exponent}$$

What is wrong: This holds for $K=2$ because the likelihood ratio $dP1/dP0$ is symmetric on the log scale (values are reciprocals). For $K > 2$, the optimal Chernoff exponent generally uses a tilting parameter $t \neq 1/2$, so $-\log(\text{rho_K}) > \text{KL}(1/2 \parallel \mu_s)$. The Hellinger exponent $-\log(\text{rho_K})$ is still a valid lower bound on the Chernoff exponent, but it is **not exact**.

Verification ($\mu=0.3$, $K=3$):

$$-\log(\text{rho_K}) = -\log(0.829) = 0.188$$

$$\text{KL}(1/2 \parallel 0.3) = 0.087$$

$$-0.188 \neq 0.087.$$

Consequence: The claim that the Hellinger approach gives “exact Chernoff exponent” is true only for $K=2$ (or more generally, when the two distributions are symmetric on the log-odds scale). For $K > 2$, the Hellinger exponent is a valid but non-exact lower bound.

How to fix: Clarify that the Chernoff exactness holds for $K=2$ and provide the general expression for $K > 2$, or simply note that the Hellinger bound is always valid and the exponent is bounded below by the $K=2$ exponent.

Issue 5 [MAJOR]: Section 9 – $C_{\text{bal}} > 1$ extension is incomplete

Location: Lines 536-619

What is wrong: The section is a sketch, not a rigorous proof. There are several gaps:

1. The convexity bound via `sqrt` concavity gives $H^2(P_0, P_1) \leq \text{avg } H^2(P_0, Q_{\text{ell}})$, which is correct. But the translation to a lower bound on the testing error uses $TV \leq H$, which yields $R \geq (1-TV)/2 \geq (1-\sqrt{\text{avg } H^2})/2$. This is a valid bound but may be very weak.
2. The analysis switches between using `max H` and `min H` inconsistently. Line 605 says “the worst (largest) Hellinger distance among the mixture components gives the tightest upper bound on TV”, but line 610 then says “the component with the smallest... gives the smallest Hellinger distance, and hence the weakest lower bound.” These are different criteria and the presentation is confusing.
3. The claim “the exponent $2M\Delta^2$ is preserved” (line 618) is asserted

without proof. While it's plausibly true (each component's gap is at least Δ times some C_{bal} -dependent factor), no rigorous derivation is given.

Consequence: The Theorem 4 statement claims to hold under (A1)-(A6) without requiring $C_{\text{bal}} = 1$, but the proof only fully handles $C_{\text{bal}} = 1$. The $C_{\text{bal}} > 1$ case is not rigorously proven.

How to fix: Either (a) add $C_{\text{bal}} = 1$ as an explicit condition in the theorem statement, or (b) provide a complete derivation for $C_{\text{bal}} > 1$. Option (b) would involve bounding the average Hellinger affinity:

$$(1/L) \sum_{\text{ell}} \rho_{\text{ell}}^M \geq \min_{\text{ell}} \rho_{\text{ell}}^M$$

where $\rho_{\text{ell}} = \sqrt{\mu_s * p_{\text{ell}}} + \sqrt{(1-\mu_s) * (1-p_{\text{ell}})}$ and $p_{\text{ell}} = 1 - C_{\text{ell}}\mu_s/(K-1)$ for some $C_{\text{ell}} \geq 1$. The smallest ρ_{ell} (closest to 1) gives the weakest bound, and this is attained at $C_{\text{ell}} = C_{\text{bal}}$ (or similar).

Issue 6 [MINOR]: Section 6.4 – Wrong Taylor coefficient for Δ^4

Location: Lines 340-344

Claim:

$$\log(1-4\Delta^2) = -4\Delta^2 - 8\Delta^4/3 - O(\Delta^6)$$

What is wrong: The Taylor expansion of $\log(1-x) = -x - x^2/2 - x^3/3 - \dots$ with $x = 4\Delta^2$ gives:

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$$\begin{aligned}\log(1-4\Delta^2) &= -4\Delta^2 - (4\Delta^2)^2/2 - (4\Delta^2)^3/3 - \dots \\ &= -4\Delta^2 - 16\Delta^4/2 - 64\Delta^6/3 - \dots \\ &= -4\Delta^2 - 8\Delta^4 - O(\Delta^6)\end{aligned}$$

The coefficient of Δ^4 is -8 , not $-8/3$.

Then $(M/2)*\log(1-4\Delta^2) = -2M*\Delta^2 - 4M*\Delta^4 - O(M*\Delta^6)$,
not $-2M*\Delta^2 - (4/3)M*\Delta^4 - O(M*\Delta^6)$.

Consequence: The specific coefficient in the $O(M*\Delta^4)$ term is wrong, but this term is absorbed into the $O(M*\Delta^4)$ notation anyway. No substantive claim is affected.

How to fix: Correct the coefficient.

3. Algebraic Verification of F1 Conversion

This is the most critical section. Below is a complete, corrected derivation of the F1 lower bound.

Setup

- Testing error: $R(\psi) = \max(\alpha, \beta)$ where $\alpha = P_0(\psi=1)$, $\beta = P_1(\psi=0)$
- Le Cam bound: $R(\psi) \geq (1-TV)/2 \geq \rho^M/4 =: \epsilon$
- So $\max(\alpha, \beta) \geq \epsilon$

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F1 in terms of alpha, beta

$$\begin{aligned} F1 &= 2*TP/(2*TP + FP + FN) \\ &= 2*\eta*(1-\beta) / (2*\eta*(1-\beta) + (1-\eta)*\alpha + \eta*\beta) \end{aligned}$$

Case analysis

Case 1: $\beta \geq \epsilon$ (false negative rate is high)

Using $FP \geq 0$:

$$\begin{aligned} F1 &\leq 2*\eta*(1-\beta) / (2*\eta*(1-\beta) + \eta*\beta) \\ &= 2*(1-\beta)/(2-\beta) \end{aligned}$$

Since $\beta \geq \epsilon$ and the function is decreasing in β :

$$\begin{aligned} F1 &\leq 2*(1-\epsilon)/(2-\epsilon) \\ 1-F1 &\geq \epsilon/(2-\epsilon) \end{aligned}$$

Case 2: $\alpha \geq \epsilon$ (false positive rate is high)

Using $FN \geq 0$:

$$F1 \leq 2*\eta*(1-\beta) / (2*\eta*(1-\beta) + (1-\eta)*\alpha)$$

Since $\beta \geq 0$ (maximizes F1 at $\beta = 0$) and $\alpha \geq \epsilon$:

$$\begin{aligned} F1 &\leq 2*\eta / (2*\eta + (1-\eta)*\epsilon) \\ 1-F1 &\geq (1-\eta)*\epsilon / (2*\eta + (1-\eta)*\epsilon) \end{aligned}$$

Universal bound

Since $\max(\alpha, \beta) \geq \epsilon$, either Case 1 or Case 2 must hold.

Therefore:

$$1-F1 \geq \min(\epsilon/(2-\epsilon), (1-\eta)\epsilon/(2\eta+(1-\eta)\epsilon))$$

Which case is tighter?

For $\eta < 1/2$ and small ϵ (the typical regime):

$$\begin{aligned} \epsilon/(2-\epsilon) & \approx \epsilon/2 \\ (1-\eta)\epsilon/(2\eta+(1-\eta)\epsilon) & \approx (1-\eta)\epsilon/(2\eta) \end{aligned}$$

Ratio: $(\epsilon/2) / ((1-\eta)\epsilon/(2\eta)) = \eta/(1-\eta) < 1$
for $\eta < 1/2$.

So Case 1 gives the smaller bound (tighter). **This contradicts the proof's claim.**

Correct F1 lower bound

$$1-F1 \geq \epsilon/(2-\epsilon) = (\rho^M/4)/(2 - \rho^M/4) = \rho^M/(8 - \rho^M)$$

For small ρ^M :

$$1-F1 \geq \rho^M/8 \quad (\text{conservative simplification})$$

Compare with the claimed bound $\rho^M/(16\eta)$:

η	Correct bound	Claimed bound	Ratio
0.1	$\rho^M/8$	$\rho^M/1.6$	5x too large

eta	Correct bound	Claimed bound	Ratio
0.3	$\rho^M/8$	$\rho^M/4.8$	1.67x too large
0.5	$\rho^M/8$	$\rho^M/8$	Match

The claimed bound is **stronger than justified** for all $\eta < 1/2$, with the discrepancy growing as η decreases.

Rate optimality preserved

Both the correct bound $\rho^M/8$ and the claimed bound $\rho^M/(16\eta)$ have leading term $\rho^M = (2\sqrt{\mu_s(1-\mu_s)})^M = \exp(-2M\Delta^2 + O(M\Delta^4))$. The rate is preserved, confirming that Part (c) is unaffected.

4. Summary of Correct vs. Claimed Results

Quantity	Claimed in v2	Correct value	Severity
Testing bound	$\rho^M/4$	$\rho^M/4$	OK
F1 bound (const)	$\rho^M/(16\eta)$	$\rho^M/(8-\rho^M)$	FATAL (const wrong)
F1 bound (rate)	$\exp(-2M\Delta^2)$	$\exp(-2M\Delta^2)$	OK

Quantity	Claimed in v2	Correct value	Severity
K>2 affinity	$2*\sqrt{\mu*(1-\mu/(K-1))}$	$\sqrt{\mu*(1-\mu/(K-1))} + \sqrt{((1-\mu)*\mu/(K-1))}$	MAJOR (formula wrong)
Chernoff exactness	Holds for all K	Holds for K=2 only	MAJOR (over-claimed)
C_bal > 1	Claimed but not proven	Needs rigorous treatment	MAJOR (gap)
Taylor coeff	$-8/3*\Delta^4$	$-8*\Delta^4$	MINOR

5. Comparison to v1: What is Fixed, What Remains

Fixed from v1

Issue	v1	v2	Status
Slud’s inequality	Fatal error (false bound)	Hellinger tensorization	Fixed
Chi-squared direction	Used wrong direction for K>2	Product reduction with correct direction	Fixed

Issue	v1	v2	Status
F1 algebra	Algebraic error in Lemma 8	Case analysis (new errors introduced)	Partially fixed, new errors
Odd M ceiling	Unaddressed	Works for all M	Fixed

Remaining issues (this review)

1. **F1 constant is wrong** (Section 7.3, 7.5) – The simplification uses backwards inequality directions. The correct bound is $1 - F1 \geq \rho^M / (8 - \rho^M)$, which is approximately $\rho^M / 8$, not $\rho^M / (16 * \eta)$.
2. **$K > 2$ affinity formula is wrong** (Section 8.3) – The formula $2 * \sqrt{\mu * (1 - \mu / (K - 1))}$ is incorrect and can give values > 1 . The correct formula is a sum of two square roots.
3. **Chernoff exactness overclaimed** (Section 11) – True for $K=2$ but not $K>2$.
4. **$C_{\text{bal}} > 1$ is a sketch** (Section 9) – Not a complete proof.

Recommendations

1. Correct the F1 bound to $1 - F1 \geq \rho^M / (8 - \rho^M)$ (or the conservative $\rho^M / 8$).
2. Correct the $K>2$ affinity formula.

3. Move the Chernoff exactness claim to the $K=2$ section only.
4. Either require $C_{\text{bal}} = 1$ explicitly in the theorem statement, or provide a complete $C_{\text{bal}} > 1$ proof.
5. Verify the corrected F1 bound still matches the numerical examples (it does: $1 - F1 \geq \rho^M/8$ gives $0.8^{10}/8 = 0.013$, which is $\leq$ the direct bound 0.027 from the full expression – this is conservative but valid).

End of review.